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A Framework to Manage Systems Based on Digital Twin

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Abstract

The integration of technologies that leads to the concept of a digital twin has received much attention in recent years. Varying definitions and use cases have been developed in this rapidly evolving field. However, the literature related to the topic has failed to come to a common understanding or unified definition. As such, the term can lead to confusion, particularly in the simulation-focused research community. In this research we focus on the use of a digital twin as a tool to support and enhance real-time decision-making. With this in mind, this talk will focus on our effort to integrate the digital twin concept with existing knowledge in decision-making and decision-making processes. In its most simplistic form, a digital twin senses/collects information from the physical environment, generates decision alternatives, runs virtual simulations in a one-to-one representation of reality to predict future states and assess alternatives, selects favorable actions, and finally actuates these decisions in the physical environment. We leverage existing frameworks of human decision-making to frame these aspects of a twin. This decision-making orientation focuses the requirements of a twin and leads to a proposed architecture based on Boyd's OODA loop. We discuss a number of open questions and technological requirements that are focused by this orientation.

Keywords:

Digital twin, simulation, real-time decision-making, OODA loop

1. Introduction

The digital twin has received much attention over the past decade, and has been theorized to revolutionize many fields. However, given its multidisciplinary nature, the digital twin concept has suffered from dilution, with the term now often being used in various settings to simply describe a virtual representation of a physical object or system. While this virtual representation might mirror the physical object in every conceivable way, its existence does not immediately characterize it as a digital twin. A virtual model of a physical system, no matter the fidelity or complexity, is still just a model. Therefore, the term digital twin refers to more than a virtual model - it specifies the *relationship* between virtual model and physical system.

The concept of "twins" dates to at least NASA's Apollo program. Engineers would build two identical space systems - one to send to space, and the other to remain on earth. The system on earth, referred to as the twin, would allow engineers to mirror the current condition and configuration of the flying twin, providing a resource to simulate alternatives [1]. The use of this twin garnered national attention during the Apollo 13 mission in 1970, where engineers had to simulate many alternative system configurations for the crew. Thankfully, the crew returned home safely, due in large part to the alternatives made available through experimentation with the twin.

In 2010, the term "digital twin" first appeared in NASA's Modeling, Simulation, Information Technology & Processing Roadmap, defining it as, "...an integrated multi-physics, multi-scale, probabilistic simulation of a vehicle or system that uses the best available physical models, sensor updates, fleet history, etc., to mirror the life of its flying twin" [2, 3]. The authors state the digital twin is capable of integrating real-time sensor updates and historical maintenance data to forecast future maintenance needs, and can provide recommendations for changes in real-time mission accomplishment.

Later, the digital twin was further defined by Grieves [4] as an innovation in product lifecycle management, and contains three main parts - physical products in real space, virtual products in virtual space, and the data connection linking the physical and virtual together. This type of relationship could be used in "front-running" a real-time system, simulating future alternatives to optimize the physical system and provide favorable scenarios to operators in real-time [5]. This is essentially how NASA used their physical twin during the Apollo 13 mission in 1970.

While the term digital twin has been around since 2010, many definitions and proposed use cases exist today. However, actual implementation is scarce. This has often been attributed to the lack of a single, unified definition [6]-[10]. This paper summarizes relevant research in the field of digital twins, then offers a framework to characterize and manage systems based on a digital twin.

2. Problem Description

Much has been published concerning digital twins and their uses. However, there is still no unified definition. As Tao, *et al* [8] states, a "unified Digital Twin modeling method is critically needed." Others ([9]-[11]) concluded most of digital twin literature did not reflect the same unified definition of a digital twin. Additionally, Jones, *et al* [9] states literature lacks a consistent view of what exactly is a digital twin, which has led to a breadth of definitions, and risks diluting the concept.

Furthermore, many authors have categorized their work as a digital twin, but what they actually described was a digital model or digital shadow [7] – concepts we describe in Section 4. According to the categorical literature review from Kritzinger, *et al* [7], only 18% of digital twin papers actually described a digital twin. Some authors describe systems nearly identical to simulation models as defined by seminal simulation texts of Banks, *et al* [12] and Law [13], but confusingly call these systems digital twins. A computer simulation can be a close approximation to an underlying physical system, with the purpose of providing a cheaper and faster way of experimentation [12, 13]. However, a computer simulation alone does not constitute a digital twin. A digital twin is more than a computer-based virtual representation of a physical system. It requires an appropriate level of integration [7], and the ability to simulate the physical system over time. While the levels of integration are established, the ability to simulate over time is not. These two characteristics, when coupled, can allow for real-time (or near real-time) decision-making. Therefore, a unified framework for digital twins should include a decision-making strategy.

3. Related Research

Digital twins are often mentioned as a decision tool for initial product development and product lifecycle management (PLM). Many researchers focus on digital twin development and use for a singular product [4, 5, 9, 10, 14]. However, literature has expanded the term to also include processes and systems. Researchers are beginning to implement digital twins within manufacturing production systems. In this setting, real-time data from sensors and actuators are used to gain understanding of the current system through simulation, thus enabling real-time decision-making [6, 15]-[17].

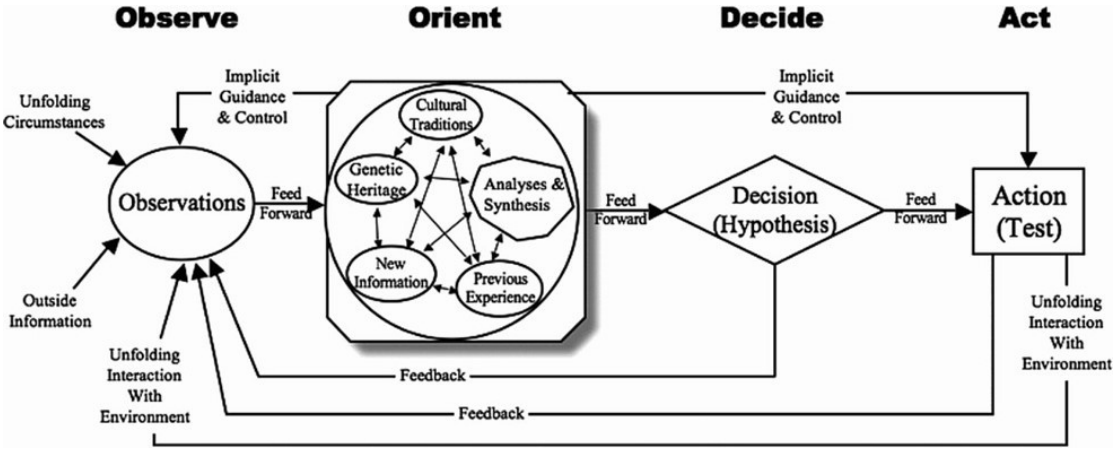
Research is also exploring implementing a digital twin specifically as a digital control system to aid in decision-making. Uhlemann, *et al* [6] describes using Industry 4.0 to allow for real-time production control of a manufacturing system, coupling commercially-available components in already isolated solutions. Stark, *et al* [17] discusses the use of digital twins in smart manufacturing, and gives examples of implementing Raspberry Pis for sensor and actuator control. Liu, *et al* [15] discusses issues with implementing a digital twin in a complex manufacturing system, including the synchronization of physical and virtual worlds, and effective feedback from decision-support information provided by optimization. Onggo [16] uses real-time data within a simulation at run-time, developing a real-time decision-support tool for manufacturing systems and defining it as a Symbiotic Simulation System (S3). Fera, *et al* [18] states the digital twin is becoming a digital controller of manufacturing systems, integrating sensors and simulations to provide near real-time management of a production line. Erkoyuncu, *et al* [19] describes the vision of a digital twin reaching a state of full autonomy, collecting information about a system's operation, simulating this system, and finally supporting any decision made about the system.

As the above research indicates, authors are using digital twins as real-time decision-making tools. However, no unified decision-making strategy exists. Some authors mention suggested steps for decision-making that are very similar to Boyd's Observe, Orient, Decide, and Act (OODA) loop. For example, according to Rosen, *et al* [1], the digital twin can observe its environment, decide on courses of action, and implement these actions in the physical environment. Other similar statements can be found in [20]-[24].

The relationship between a digital twin and the OODA loop is mentioned in Curry [25], with the author stating, "at the heart of a Digital Twin is the OODA loop." Specifically, the author details each of the four OODA steps as they

apply to creating dashboards for smart energy and water systems. The observation step is supported by a Real-time Linked Dataspace (RLD), which is a collection of various data sources. The orientation step supports situational awareness via dashboards and makes use of real-time and historical data to improve a user's experience. A dashboard showcasing water usage, conservation tips, and ways to improve consumption are then displayed for use by decision-makers. Within the decision step, the author suggests employing a digital twin to detect stagnant water. Finally, the act step communicates alerts and tasks to the user for implementation. However, the author focuses more on the RLD as an enabler of smart environments and less on the actual classification and construction of a digital twin. Human interaction is also required at each of the orientation, decision, and action steps. This paper intends to unify and further define the definition of a digital twin, and unlike the description of Curry [25], we claim a digital twin is not simply the decision component of the OODA loop, but at its most fundamental level, a digital twin *is* a digital OODA loop.

Colonel John Boyd was a United States Air Force officer and fighter pilot. In dogfights, Boyd was known as "40-second Boyd", as he was reportedly able to defeat his enemy in 40 seconds or less. Boyd later described his winning strategy as a simple loop: *observe* the surroundings, *orient* to the surroundings, *decide* on an action, then *act*. This methodology, known as the OODA loop, provided Boyd the ability to adjust to a rapidly-changing environment faster than his opponents [26]. Boyd's OODA loop is shown in **Figure 1**.



4. Methodology

All digital twins are constructed to make a decision. Therefore, a digital twin should follow a decision-making process. We use Boyd's OODA loop, due to its clear and straightforward representation of how a decision can be made, and to its components appearing in the literature. As mentioned in Section 3, Boyd developed the OODA loop to illustrate the decision-making process. If we apply the OODA loop as a methodology for defining a digital twin, we can begin constructing a framework. The following is a description of a digital twin as a digital implementation of the OODA loop:

Observe We observe our physical environment and system using a variety of sensors and tags. Observations are then automatically available to the virtual environment, where models can use this information to generate courses of action.

Orient Given the current state of the physical system, we build and implement virtual models with the specific purpose of predicting future states and assessing alternatives. Boyd considered this the most important step.

Decide Considering the output of our simulation, we decide what to do next.

Act If the result of the decision step is an action, we implement the decision in the physical system. Once the action is complete, we begin the loop again, starting with the Observe step.

Using the four steps of the OODA loop, the required key components of a digital twin are listed in **Table 1**. Additionally, we propose a digital twin framework in **Figure 2**.

Table 1: Digital Twin Components

Observe	to sense, collect, and communicate the current state of the physical system
Orient	to predict future states and alternatives, simulating the system over time
Decide	to select favorable actions
Act	to communicate and actuate these actions in the physical system

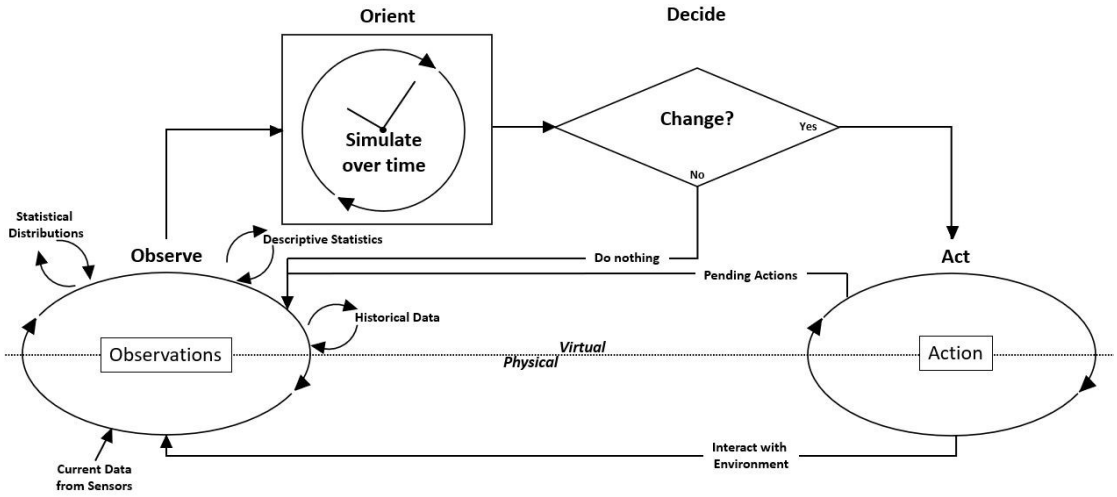


Figure 2: Digital Twin Framework (adapted from [26])

With the four key components of a digital twin defined, we can now begin to specify their function. Specifically, both the Observe and Act components must be capable of automatic bidirectional communication, and the Orient step (the model) must simulate the underlying physical system over time.

4.1. Automatic Bidirectional Communication

According to Kritzinger, *et al* [7], bidirectional communication between the physical and virtual (and how the bidirectional communication is achieved) is a key factor in characterizing a digital twin, specifying three levels of integration:

Digital Model Both physical and virtual objects communicate via manual data flow [7]. This is likely achieved through human input, where the user records observations from the physical environment, updates a virtual model, then makes changes to the physical object based on the model's output.

Digital Shadow Similar to a digital model, the digital shadow has automatic data flow from physical to virtual, but still maintains manual data flow from virtual to physical [7]. In this case, the user has implemented automatic data collection and communication. Thus, physical system data flows to the virtual model automatically.

Digital Twin Both the physical-to-virtual communication and virtual-to-physical communication are automatic. Here, the physical object automatically updates the virtual, and the virtual object automatically changes the physical [7]. Once this bidirectional communication is automatic, the user no longer has an active role in virtual model execution

or physical system changes. The system can now self-monitor and adapt, implementing changes to the physical system when the virtual determines a change is beneficial.

We can now use our four digital twin components to further clarify the levels of integration from Kritzinger, *et al* [7], as shown in **Table 2**.

Table 2: Levels of Integration [7] and Digital Twin Components

	Digital Model	**Digital Shadow**	**Digital Twin**
Observe	Manual	Automatic	Automatic
Orient	Automatic	Automatic	Automatic
Decide	Manual	Manual/Automatic	Automatic
Act	Manual	Manual	Automatic

Here, we assume the Orient step is an already-constructed simulation model that accepts inputs from the physical system and predicts future states and alternatives, where the running of this model is considered automatic.

Once the level of digital twin integration is achieved, the virtual model has complete control over decisions in the physical system. Therefore, further explanation is needed to clarify what exactly is required from the virtual model. Specifically, in addition to the levels of integration from Kritzinger, *et al* [7], we propose a digital twin must also simulate the physical system over time.

4.2. Simulate Over Time

According to Banks, *et al* [12], "A simulation is the imitation of the operation of a real-world process or system over time." Simply creating a realistic model of a system is not enough. We must also consider how this underlying physical system behaves over time. Whether deciding when vehicle maintenance should occur, estimating the remaining life of an aircraft wing, or determining the optimal sequence in a manufacturing system, these activities require simulating future outcomes.

Simulation modeling can be used to understand and predict the effect of changes to existing systems [12]. Therefore, simulating over time is a key characteristic of a digital twin (as well as a digital model and digital shadow). Deciding which level of integration from Kritzinger, *et al* [7] to use will likely be determined by how quickly a decision is needed, and how much control over the physical system the virtual is desired to have.

5. Conclusion

The digital twin concept has suffered from dilution since its inception. As researchers continue to theorize digital twin use in their respective fields, this dilution is likely to continue. A simple, unified framework is key to standardize research and use-cases across all fields. As digital twins are decision-making tools, a framework modeled after a decision-making strategy is paramount. Our research indicates many authors already use at least one of the four digital twin components listed in this paper. We hope that combining these components into a framework will assist in a common understanding of the digital twin concept.

Disclaimer

The views expressed in this article are those of the authors and do not reflect the official policy or position of the United States Air Force, the United States Department of Defense, or the United States Government.

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